

Early Detection of Peripheral IV Infiltration/Extravasation

Using Intelligent Thermal Video Analytics

A Smartphone–Thermal Camera–Based Nursing Assistant

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Motivation: Why Early Detection Matters

- Late detection → tissue injury, longer treatment, higher cost
- Usually detected after visible signs (swelling, erythema)
- Visual inspection by nurses → subjective and delayed
- Higher-risk groups: chemotherapy patients; older adults (reduced venous elasticity)

Specific Objectives

- ROI-based pipeline: auto-identify IV site (phone-mounted thermal camera)
- Lightweight smartphone model: early risk score from 16–32 frame clips
- Validate using ethics-approved real-world clinical data
- Offline mobile deployment (no internet)

Clip Extraction, Inputs, and Labeling

Clip extraction and inputs:

- Short clips of 16–32 frames (~ 2 –4 seconds), ROI stabilization, and micro-motion correction.
- Three-channel input: raw temperature (T), temporally smoothed T , and ΔT relative to individual baseline.
- Model input resolution: 112×112 .

Labeling (public clips, development phase):

- Positive if a temperature drop is observed within 0–120 seconds after the end of a clip.
- Negative if no temperature difference is observed for at least 5 minutes.
- Exclude samples with ambiguous timestamps.

Architectures:

- Primary: TSM-MobileNetV2.
- Baselines: R(2+1)D and S3D.

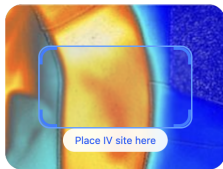
Mobile deployment:

- INT8 quantization and on-device deployment using TensorFlow Lite or ONNX Runtime Mobile.
- Targets: end-to-end latency ≤ 2 seconds; offline operation; model size ≤ 10 MB.

Proposed Nursing Workflow



Align camera with IV site



- ✓ ROI centered
- ✓ Patient ID confirmed
- ✓ Privacy ok

Start monitoring

9:41

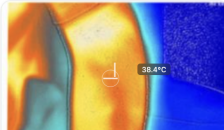


Sarah Mitchell

Age 42 • Bed 204-A

Thermal Image

Updated 2s ago



Normal Elevated High



Infiltration suspected

Risk score: 0.87 – high

Alarm condition met: score > threshold
for 10+ seconds

Abnormal temperature pattern around IV
site before visible swelling.

✓ Mark as checked

🕒 Snooze 10 min

9:41



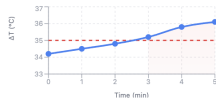
Alarm details Infiltration suspected

JD

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42 y.o. • Bed 204-A

IV site ROI

ΔT temperature trend (last 5 min)



- Threshold exceeded for 10+ seconds
- Abnormal warming pattern around IV site
- Early warning ~2 min before visible signs

Mark as checked

Thank you
Questions?